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The Effect of Boron on the Corrosion Resistance of the High Entropy Alloys $Al_{0.5}CoCrCuFeNiB_x$

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High entropy alloys are a newly developed family of multicomponent alloys that consist of various major alloying elements, including copper, nickel, aluminum, cobalt, chromium, iron, and others. Each element in the alloy system is present at between 5 and 35 atom %. A high entropy alloy has numerous beneficial mechanical, magnetic, and electrochemical characteristics. This investigation discusses the corrosion resistance of the $Al_{0.5}CoCrCuFeNiB_x$ alloys with various amounts of added boron. Surface morphological and chemical analyses verified that the addition of boron produced Cr, Fe, and Co borides. Therefore, the fraction of Cr outside borides precipitates was scant. The anodic polarization curves and electrochemical impedance spectra of the $Al_{0.5}CoCrCuFeNiB_x$ alloys, obtained in 1 N H_2SO_4 aqueous solution, clearly reveal that the general corrosion resistance decreases as the concentration of boron increases.
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For years, the development of traditional alloy systems has been based primarily on one principal element, such as iron-based, aluminum-based, and magnesium-based alloys and nickel-based superalloys.[1] Since the 1970s, intermetallic compounds of Ti–Al, Ni–Al, and Fe–Al binary systems have attracted considerable attention because they have extremely high specific strength and thermal resistance.[2] In the 1990s, Inoue et al.[3] obtained the highly glass-forming alloy by using at least three different elements with significantly different atomic radii. In 1993, Greer[4] came up with the "confusion principle" in which he stated that the possibility of crystallization could be lowered if the number of alloying elements was large. However, the designs of the aforementioned alloys are still limited in that their matrices always contain one or two major elements. A novel high entropy alloy was developed herein. High entropy alloys are composed of five or more principal elements, each at 5 to 35 atom %. This study focuses on a new alloy design with multiple principal elements in equimolar or near-equimolar ratios.[5,6]

Solid solutions with more principal elements tend to be more stable because they have larger mixing entropies. The configurational entropy change per mole, ΔS_{conf}, during the formation of a solid solution from seven elements with equimolar fractions may be calculated from the following equation

$$\Delta S_{conf} = k \ln \omega = -R\sum_{i=1}^{7} x_i \ln x_i = R \ln 7 \approx 1.95R, \quad x_i = \frac{1}{7} \qquad [1]$$

where k is the Boltzmann's constant, ω is the number of ways of mixing, and R is the gas constant, 8.314 J/K mol. For instance, ΔS_{conf} of equimolar alloys with 3,5,7, and 9 elements are 1.10 R, 1.61 R, 1.95 R, and 2.20 R, respectively. In fact, when other positive vibrational, electronic, and magnetic contributions are considered, the entropy change associated with mixing in equimolar high entropy alloy is greater than the calculated one.[7]

Extensive trials have yielded various alloy systems with simple crystal structures and extreme properties. In our previous studies,[8,9] the $Al_xCoCrCuFeNi$ alloys with various aluminum contents (x values in the molar ratio, $x = 0$–1.0) were obtained by the casting method. These alloys possessed simple face-centered cubic/body-centered cubic (fcc/bcc) structures. Alloys with aluminum contents of less than 0.5 mol are composed of a simple fcc solid-solution structure. When the aluminum content is 0.8 mol, a bcc structure appears and constructs with mixed fcc and bcc eutectic-phases. Spinodal decomposition occurs when the aluminum content exceeds 1.0 mol. The $Al_{0.5}CoCrCuFeNi$ alloy is ductile, can be effectively work-hardened, and is strong at temperatures up to 800°C, indicating that this alloy has potential applications in high-temperature structures and working tools. Adding boron to the $Al_{0.5}CoCrCuFeNi$ alloy increases the hardness and wear resistance by forming the borides. For example, the $Al_{0.5}CoCrCuFeNiB_x$ alloy has an excellent hardness of Hv 736, and a wear resistance of 1.76 m/mm^3 is even higher than that of SUJ bearing (1.52 m/mm^3).[9] However, the corrosion characteristics of the $Al_{0.5}CoCrCuFeNiB_x$ alloys in most aqueous solutions are unknown. The purpose of this study was to investigate the electrochemical behavior of the $Al_{0.5}CoCrCuFeNiB_x$ alloys in 1 N sulfuric acid.

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Experimental

Test materials.— Elements Al, Co, Cr, Cu, Fe, and Ni with purities of over 99 wt % were present as granules as raw materials. Fe–18.2 wt % B master alloy was used with the added boron. This high entropy alloy is a seven-component alloy in the form of $Al_{0.5}CoCrCuFeNiB_x$, with 0.5 mol of aluminum, and 1 mol of the other five alloying elements and between zero and 1 mol of the added boron. Table I presents the compositions of the alloys. The multiprincipal-element $Al_{0.5}CoCrCuFeNiB_x$ alloy system with various boron contents with x from 0 to 1.0 was prepared in this work. An induction furnace was used to melt the high entropy alloys at 1600°C after purging with argon. First, raw materials of Fe and Ni were placed in the crucible. When Fe and Ni had melted in liquid state, then Cr, Co, FeB, Cu, and Al were added to the crucible in order according to their melting points. Finally, the melt was poured into a shell mold preheated at 1100°C. The size of the ingots was $8 \times 7 \times 14$ (cm). The high entropy alloy cylinder for electrochemical measurements was obtained by cutting the bulk material by electric arc line into pieces that were 3 mm thick and 8 mm in diameter. A 304 stainless steel rod was machined down to 5 mm thick and 12.7 mm in diameter. Each test specimen was then cold-mounted, using an epoxy resin to give an exposed surface area of 0.5 cm^2 for the high entropy alloys and 1.26 cm^2 for the 304 stain-

Table I. High entropy alloys used in this experiment.

Designation	Composition (atomic mole ratio)	Boron (atom %)
B 0	$Al_{0.5}CoCrCuFeNi$	0
B 0.2	$Al_{0.5}CoCrCuFeNiB_{0.2}$	3.5
B 0.6	$Al_{0.5}CoCrCuFeNiB_{0.6}$	9.8
B 1.0	$Al_{0.5}CoCrCuFeNiB_{1.0}$	15.4

less steel. Before each electrochemical experiment, all specimens were mechanically polished using a series of 240–600 SiC grit papers and cleaned with acetone and distilled water.

Electrochemical measurements.— Test solutions were conducted in 1 N H_2SO_4 solution at 25°C under atmospheric pressure. The test solution was deaerated by bubbling purified nitrogen gas before and throughout the electrochemical tests, to eliminate the effect of dissolved oxygen. Both electrochemical polarization and electrochemical impedance spectroscopy (EIS) were performed in a typical three-electrode cell that consisted of a specimen as the working electrode, an Ag/AgCl in 3 M KCl as the reference electrode, and a Pt sheet as the counter electrode. The Ag/AgCl scale (3 M KCl) is 208 mV lower than the standard hydrogen electrode (SHE)

$$V_{Ag/AgCl} + 0.208\ mV = V^{10}_{SHE} \qquad [2]$$

Anodic polarization measurements were made at a scan rate of 1 mV/s from an initial potential of −0.8 V to a final potential of 1.5 V vs the open-current potential (OCP). The potential was controlled and the current was measured using a potentiostat (Autolab PGSTAT30). The EIS were recorded in a range of frequencies from 10 kHz to 10 mHz. The amplitude of sinusoidal signals was 10 mV around the OCP, as determined using an Autolab PGSTAT30/FRA system from ECO Chemie. Before either of these tests was conducted, the OCP was recorded for 15 min to yield a steady-state potential. Then, the specimen was cathodically polarized to a potential of −192 m V_{SHE} for 5 min to reduce the existing surface films.

Surface morphology and chemical analysis.— Following the polarization experiment, the specimen was cleaned using distilled water and then dried in nitrogen. Immediately thereafter, the morphology and chemical composition of the corroded surface of the high entropy alloy was investigated using a JEOL 6500 scanning electron microscope (SEM) and energy-dispersive spectrometry (EDS). The chemical structure of the high entropy alloy after the polarization experiment was examined by X-ray photoelectron spectroscopy (XPS, Perkin-Elmer model PHI 1600) using a single Mg Kα X-ray operating at 250 W. Energy calibration was done using the Au $4f_{7/2}$ peak at 83.8 eV, and the energy resolution was 0.2 eV for the core-level spectra. The following XPS regions were examined for Cr 2p, Co 2p, and Fe 2p.

Results and Discussion

Anodic polarization curves.— A wealth of information is available on the corrosion behaviors of the 304 stainless steel that is exposed to H_2SO_4 solutions.[11-18] Hence, comparing the corrosion behaviors of high entropy alloys and conventional ferrous alloy,[19] such as 304 stainless steel, is of interest. Figure 1 shows that the corrosion potential (E_{corr}) of the $Al_{0.5}$CoCrCuFeNi high entropy alloy (−0.080 V_{SHE}) is apparently more noble than that of 304 stainless steel (−0.151 V_{SHE}), and the corrosion current density (i_{corr}) of the $Al_{0.5}$CoCrCuFeNi high entropy alloy (3.19 μA/cm^2) is also lower than that of the 304 stainless steel (33.18 μA/cm^2) by an order of magnitude in 1 N H_2SO_4 solution. Additionally, the 304 stainless steel has a wider region of the passive potential than the $Al_{0.5}$CoCrCuFeNi high entropy alloy. Clearly, the $Al_{0.5}$CoCrCuFeNi alloy is more resistant to general corrosion than the 304 stainless steel (higher E_{corr}, lower i_{corr}) in 1 N H_2SO_4 solution.

Cyclic potentiodynamic polarization.— Clearly, the 304 stainless steel does not suffer from pitting corrosion in halide-free solution.[10] A cyclic polarization technique was adopted to determine whether the $Al_{0.5}$CoCrCuFeNiB$_x$ alloy (x = 0, 0.2, 0.6, and 1.0) behaves similarly in H_2SO_4 solution at approximately 25°C. The hysteresis revealed by the cyclic polarization curve provides information on the pitting corrosion rate and how readily a passive film repairs itself. Negative hysteresis occurs when the reverse scan current density is less than that for the forward scan, and positive hysteresis occurs when the reverse scan current density exceeds the forward scan current density.[20]

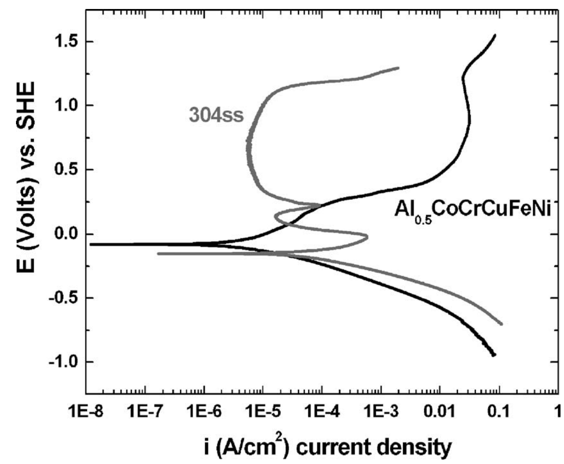

Figure 1. Comparisons of the anodic polarization curves for the $Al_{0.5}$CoCrCuFeNi alloy and the 304 stainless steel in deaerated 1 N H_2SO_4.

Cyclic polarization measurements were made at a scanning rate of 10 mV/s. The potential scan began at −0.8 V_{SHE} and continued in the anodic direction until the potential was 1.2 V_{SHE}, at which value the potential scan was reversed, and returned to where the polarization began. Figure 2 plots the cyclic polarization curves of the $Al_{0.5}$CoCrCuFeNiB$_x$ alloys with various boron contents (x = 0, 0.2, 0.6, and 1.0). The black and gray arrows next to the forward and reverse anodic branches indicate the potential scan directions. The corresponding E_{corr} and i_{corr} for the $Al_{0.5}$CoCrCuFeNiB$_x$ (x = 0, 0.2, 0.6, and 1.0) alloys vary from −115 to −159 mV$_{SHE}$, and from 787 to 2848 μA/cm^2, respectively. Table II presents the electrochemical parameters of the $Al_{0.5}$CoCrCuFeNiB$_x$ alloys (x = 0, 0.2, 0.6, and 1.0) in deaerated 1 N H_2SO_4 solution. The $Al_{0.5}$CoCrCuFeNi alloy has a higher E_{corr} and lower i_{corr} than the $Al_{0.5}$CoCrCuFeNiB$_x$ alloy (x = 0.2, 0.6, and 1.0). In our previous study,[9] the as-cast $Al_{0.5}$CoCrCuFeNi high entropy alloy was composed of a simple fcc solid-solution structure of the XRD patterns as shown in Fig. 3. However, as boron content increases in the alloys, the peak intensities of Cr and Fe borides increase as well, because of the strong bonding with boron. The cyclic polarization curve of the $Al_{0.5}$CoCrCuFeNi and $Al_{0.5}$CoCrCuFeNiB$_{0.2}$ alloy are referred to as a pseudo-passive curve because the curve does not exhibit a distinct primary passivation potential (E_{pp}), but a breakdown potential (E_b) and a transpassive region are observed.[20,21] The method for qualitatively estimating E_b and E_{pp} potential is to extrapolate the linear portions of the active, passive, and transpassive regions. The difference between E_b and E_{pp} is the estimated passive region width (ΔE). The corresponding passive region of $Al_{0.5}$CoCrCuFeNi and $Al_{0.5}$CoCrCuFeNiB$_{0.2}$ alloy are 273 and 256 mV$_{SHE}$, respectively. However, the passive regions of the $Al_{0.5}$CoCrCuFeNiB$_{0.6}$ and $Al_{0.5}$CoCrCuFeNiB$_{1.0}$ alloy totally disappeared and referred to the active curve due to the increasing effect of boron. The negative hysteresis in the reverse scans of cyclic polarization curves indicates that the $Al_{0.5}$CoCrCuFeNiB$_x$ alloys (x = 0, 0.2, 0.6, and 1.0) are not susceptible to localized corrosion in 1 N H_2SO_4.

A passive film breaks down when the applied potential is raised into the transpassive region of an anodic polarization curve, and the formation of pits can be initiated at discrete locations on the metal surface.[22] In Table II the repassivation potential (E_{rp}) is more noble than the E_{corr}, regardless of whether boron is incorporated. A passive film is generally believed to repair itself when the repassivation potential exceeds the corrosion potential. As the boron content in the $Al_{0.5}$CoCrCuFeNiB$_x$ alloys increases from 0 to 1.0 mol, the E_{rp} falls

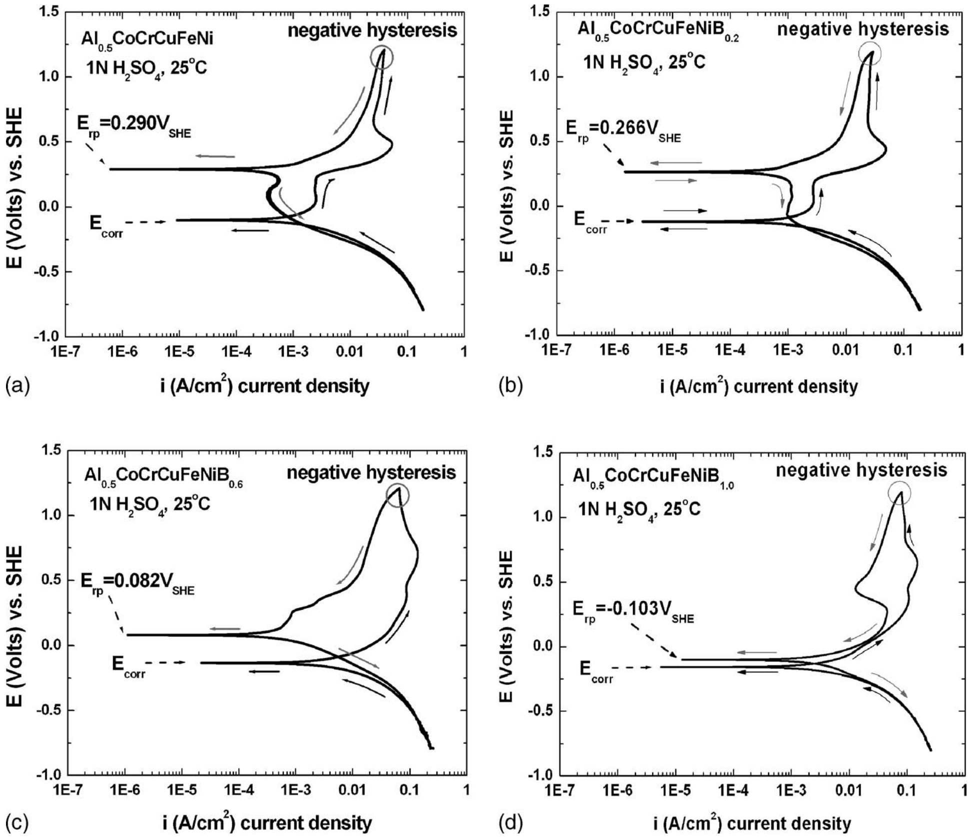

Figure 2. Cyclic polarization curves of the $Al_{0.5}CoCrCuFeNiB_x$ alloys with various boron (a) $x = 0$, (b) $x = 0.2$, (c) $x = 0.6$, and (d) $x = 1.0$ mol in deaerated 1 N H_2SO_4.

Table II. Electrochemical parameters of the $Al_{0.5}CoCrCuFeNiB_x$ alloys in deaerated 1 N H_2SO_4 solution.

Designation	E_{corr} (mV_{SHE})	E_b (mV_{SHE})	ΔE (mV_{SHE})	E_{rp} (mV_{SHE})	i_{corr} ($\mu A/cm^2$)	R_p ($\Omega\ cm^2$)
$Al_{0.5}CoCrCuFeNi$	−115	233	273	290	787	6652
$Al_{0.5}CoCrCuFeNiB_{0.2}$	−121	215	256	266	1025	2763
$Al_{0.5}CoCrCuFeNiB_{0.6}$	−148	–	–	82	2626	1872
$Al_{0.5}CoCrCuFeNiB_{1.0}$	−159	–	–	−103	2848	1081

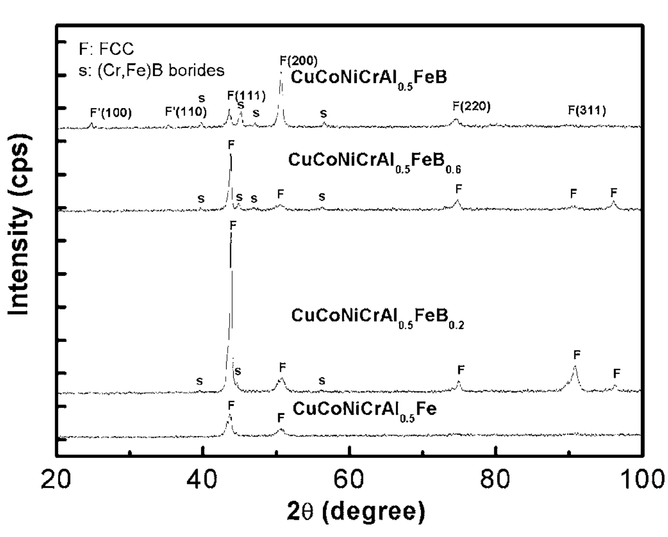

Figure 3. XRD patterns for as-cast $Al_{0.5}CoCrCuFeNiB_x$ high entropy alloys.[9] Note that "F′" peaks at 25° and 35° for the $Al_{0.5}CoCrCuFeNiB$ alloy indicate the existence of an ordered fcc.

from 0.290 to −0.103 V_{SHE}. Adding boron to $Al_{0.5}CoCrCuFeNiB_x$ alloys reduces the ability to restore its passive film.

SEM photomicrographs of corroded surfaces.— Figure 4a presents the SEM microstructure of the boron-free $Al_{0.5}CoCrCuFeNi$ alloy after anodic polarization in deaerated 1 N H_2SO_4; the cast structure is composed of dendrites and interdendritic areas. The corrosion of the $Al_{0.5}CoCrCuFeNi$ alloy took place mainly in the interdendritic regions. According to a previous report,[23] the interdendrite phase of the $Al_{0.5}CoCrCuFeNi$ alloy is a Cu-rich phase, and the dendrite is a Cu-depleted phase. As the Cu concentration declines in the high entropy alloys, the Cu segregated in interdendrite region is gradually removed. Figures 4b-d display the SEM microstructures of the $Al_{0.5}CoCrCuFeNiB_x$ alloys (x = 0.2, 0.6, and 1.0), and show string-shaped precipitates on the surface when adding boron to the $Al_{0.5}CoCrCuFeNi$ alloy. Notably, the volume fraction of stringlike precipitates increases with the amount of boron in the alloy. The corrosion on the $Al_{0.5}CoCrCuFeNiB_x$ (x = 0.2–1.0) alloy was concentrated close to the stringlike precipitates.

Analysis of surface composition by EDS.— Figure 5b presents an X-ray mapping of chromium in the $Al_{0.5}CoCrCuFeNiB_{0.6}$ alloy following the anodic polarization in deaerated 1 N H_2SO_4. Clearly, the stringlike precipitates are, for example, rich in Cr. To verify the composition of the stringlike phase, the EDS analysis was used, and

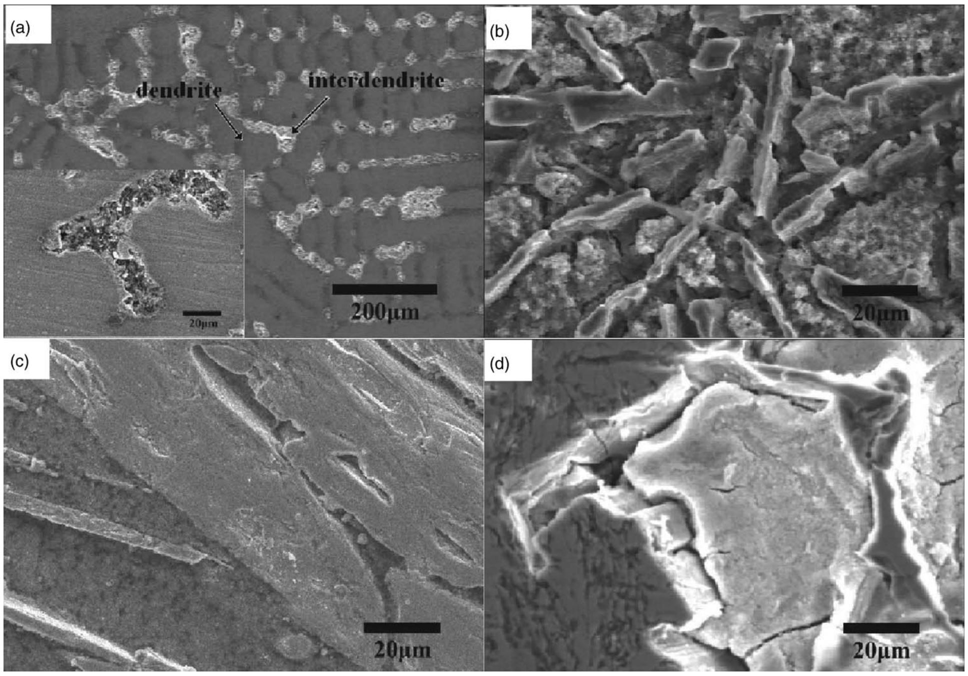

Figure 4. SEM image of the $Al_{0.5}CoCrCuFeNiB_x$ alloys with various boron (a) $x = 0$, (b) $x = 0.2$, (c) $x = 0.6$, and (d) $x = 1.0$ mol after anodic polarization in deaerated 1 N H_2SO_4.

the result is listed in Table III. There are two areas, designated A and B in Fig. 5a after the anodic polarization of the alloy $Al_{0.5}CoCrCuFeNiB_{0.6}$ in deaerated 1 N H_2SO_4. Area A is the stringy precipitate and area B is the region beyond the precipitate. EDS analysis demonstrates that the stringy precipitates of the $Al_{0.5}CoCrCuFeNiB_{0.6}$ alloy are rich in Cr, Fe, Co, and B, but poor in Cu, Ni, and Al. The atomic percentage of Cr in the stringy precipitates (50.06%) considerably exceeds that in the other region (2.48%) in the $Al_{0.5}CoCrCuFeNiB_{0.6}$ alloy. The localized attack is due to the differential compositions, which would eventually lead to the operation of galvanic cells appearing on the side of the borides precipitates. EDS analysis and polarization data revealed that the adding of boron produced Cr, Fe, and Co borides, which would apparently impair the corrosion resistance of the $Al_{0.5}CoCrCuFeNiB_x$ alloy.

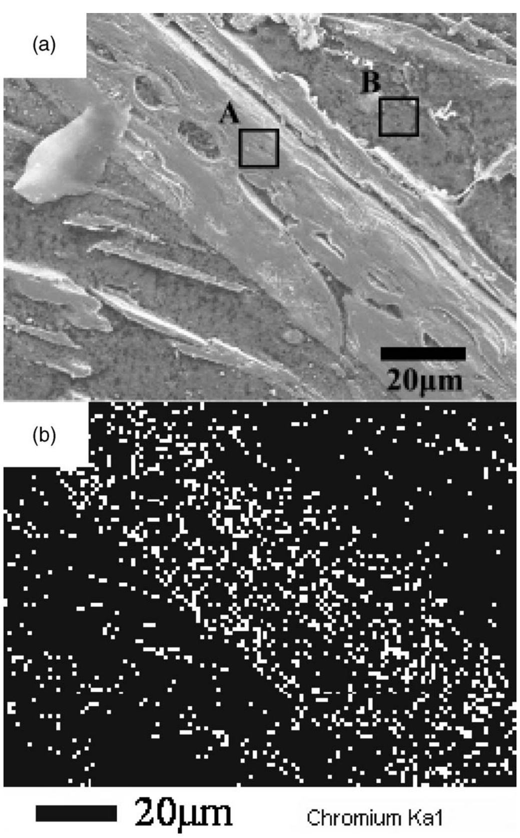

Figure 5. (a) Secondary electron image and (b) X-ray mapping of chromium in the $Al_{0.5}CoCrCuFeNiB_{0.6}$ alloy following the anodic polarization in deaerated 1 N H_2SO_4.

Chemical analysis by XPS.— The chemical composition of the high entropy alloys $Al_{0.5}CoCrCuFeNiB_x$ ($x = 0$ and 0.6 mol) was investigated by XPS using various Ar-ion sputter times. Figure 6 shows the Cr 2p core-level spectra of the $Al_{0.5}CoCrCuFeNi$ and $Al_{0.5}CoCrCuFeNiB_{0.6}$ alloy following the anodic polarization in deaerated 1 N H_2SO_4 solution. Whether sputtered or not, the binding energy position of Cr_2O_3 always remained at 576.9 eV for the boron-free $Al_{0.5}CoCrCuFeNi$ alloy.[24-26] However, the peak of Cr_2O_3 at 576.9 eV declined in intensity as the sputter time of the $Al_{0.5}CoCrCuFeNiB_{0.6}$ alloy increased. Note that the addition of boron leads to the peak of CrB_2 bonding (574.3 eV)[24] in the boron-containing $Al_{0.5}CoCrCuFeNiB_{0.6}$ alloy using Ar-ion sputter for 30 and 60 s. In 1911, Monnartz[27] first noted the outstanding passivity

Table III. EDS results for the $Al_{0.5}CoCrCuFeNiB_{0.6}$ alloy with different selected area A and B after the anodic polarization in deaerated 1 N H_2SO_4.[a]

	A		B	
Element	wt %	atom %	wt %	atom %
Al K	0.32	0.53	11.18	21.73
Co K	11.87	9.05	11.56	10.28
Cr K	63.71	55.06	2.46	2.48
Cu L	—	—	23.87	19.70
Fe K	14.85	11.95	7.02	6.59
Ni K	4.43	3.39	43.91	39.22
B K	4.82	20.02	—	—
Totals	100.00		100.00	

[a] Reference to Fig. 5a.

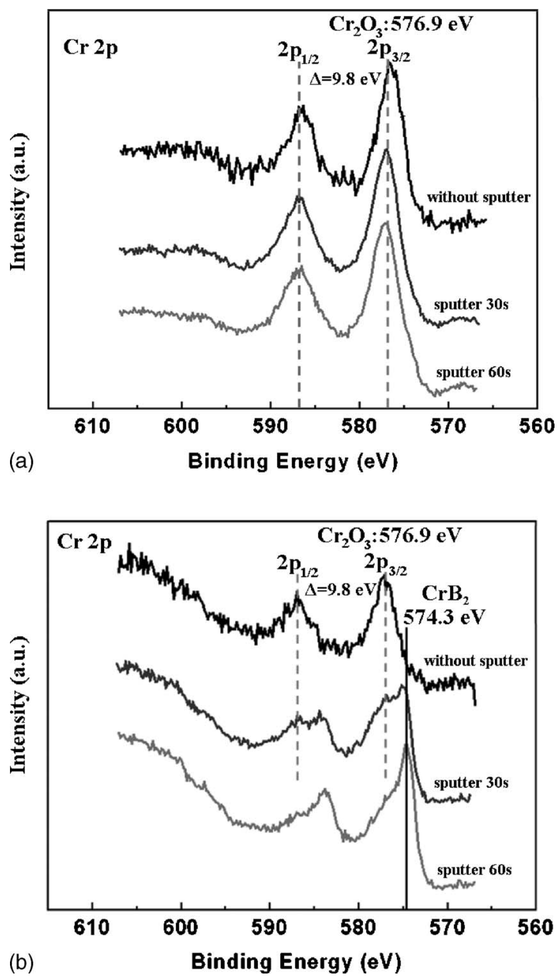

Figure 6. Typical Cr (2p) core-level XPS spectra of (a) $Al_{0.5}CoCrCuFeNi$ and (b) $Al_{0.5}CoCrCuFeNiB_{0.6}$ alloy after the anodic polarization in deaerated 1 N H_2SO_4 using various Ar-ion sputter times of 0, 30, and 60 s.

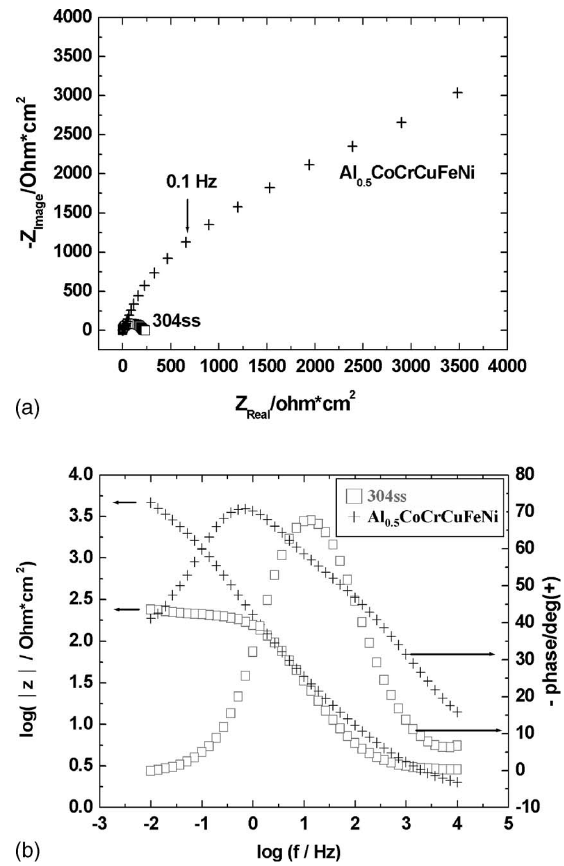

Figure 7. EIS spectra of the $Al_{0.5}CoCrCuFeNi$ alloy (+) and the 304 stainless steel (□) in deaerated 1 N H_2SO_4. (a) Nyquist and (b) Bode plots.

of stainless steel that contained at least 12% Cr. The relevant studies[28] also suggested that Cr_2O_3 is the main passivating compound on the stainless steel. The results of XPS analysis indicated the production of CrB_2 for the boron-containing alloy is reducing the Cr_2O_3 passive film on the alloy surface, e.g., $Al_{0.5}CoCrCuFeNiB_{0.6}$. Furthermore, the Co 2p and Fe 2p XPS spectra show that the bonding of Co_2B (778.4 eV) and FeB (707.1 eV)[24] are present on the surface of the $Al_{0.5}CoCrCuFeNiB_{0.6}$ alloy after 30 s of Ar-ion sputtering.

Electrochemical impedance spectroscopy.— As shown in Fig. 7a, the Nyquist plot for the $Al_{0.5}CoCrCuFeNi$ alloy presents a long tail in the low-frequency region, while the 304 stainless steel has only one depressed semicircle. The diameter of semicircle for the $Al_{0.5}CoCrCuFeNi$ alloy considerably exceeds that of the 304 stainless steel. The corrosion resistance values of 304 stainless steel and high entropy alloy were estimated by the Nyquist curves. The solution resistance (R_s) was estimated from the impedance at high frequency, while the sum of R_s and the polarization resistance R_p was estimated from the impedance at low frequency. The difference between these two impedance values is R_p, which is inversely proportional to the corrosion rate.[29,30] The corresponding Bode impedance plots, shown in Fig. 7b, show that the impedance of the $Al_{0.5}CoCrCuFeNi$ alloy greatly exceeds that of the 304 stainless steel. Therefore, the corrosion rate of the $Al_{0.5}CoCrCuFeNi$ alloy is much smaller than that of the 304 stainless steel. This result is consistent with the anodic polarization curves, as shown in Fig. 1 at the open current potential.

Figure 8 shows the effect of boron on the Nyquist plot of the $Al_{0.5}CoCrCuFeNiB_x$ ($x = 0$, 0.2, 0.6, and 1.0) alloys in deaerated 1 N H_2SO_4. The impedance of the $Al_{0.5}CoCrCuFeNiB_x$ alloy decreases as the alloying boron increases. Therefore, these tendencies in EIS measurements are consistent with the cyclic polarization curves as shown in Fig. 2. All of the $Al_{0.5}CoCrCuFeNiB_x$ alloys curves present a long tail at low frequency. A diffusion tail is often

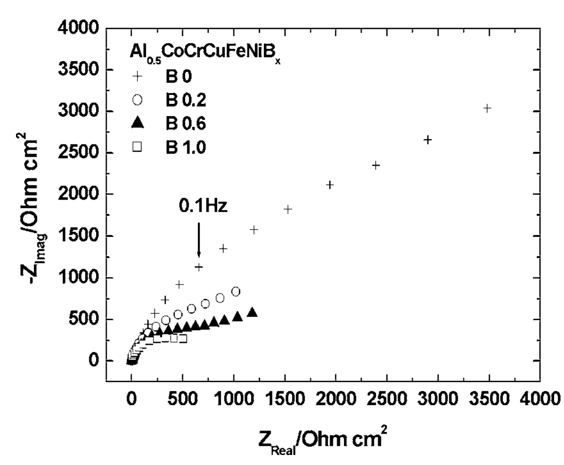

Figure 8. The effect of boron on the Nyquist plot of the $Al_{0.5}CoCrCuFeNiB_x$ ($x = 0$, 0.2, 0.6, and 1.0) alloys in deaerated 1 N H_2SO_4.

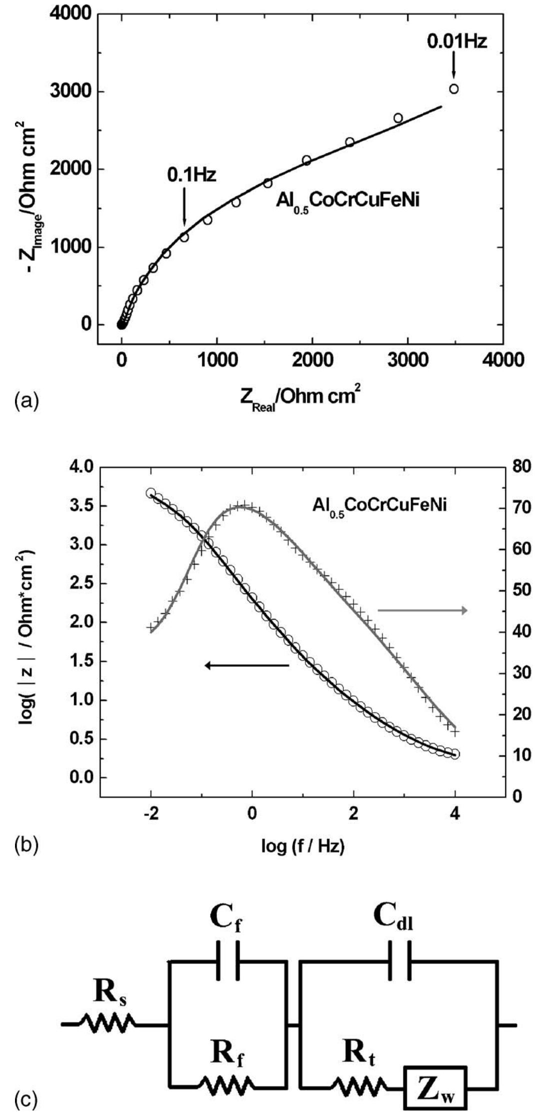

Figure 9. Experimental data (◦ and +) and simulated (solid line) of (a) Nyquist plot, (b) Bode plots, and (c) the equivalent electrical circuit representative of the electrode interface for the $Al_{0.5}CoCrCuFeNi$ alloy in deaerated 1 N H_2SO_4 solution.

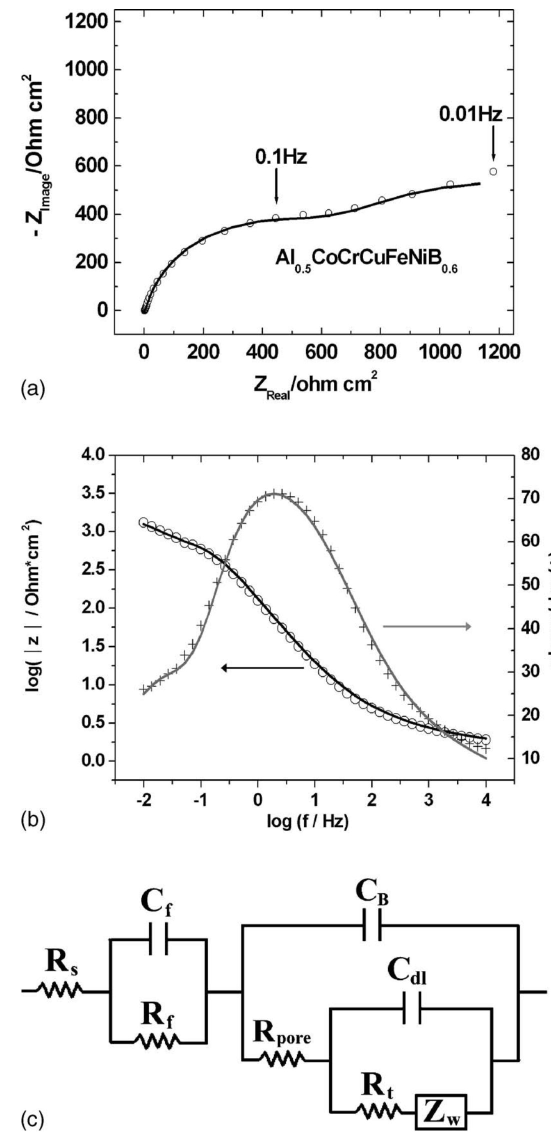

Figure 10. Experimental data (◦ and +) and simulated (solid line) of (a) Nyquist plot, (b) Bode plots, and (c) the equivalent electrical circuit representative of the electrode interface for the $Al_{0.5}CoCrCuFeNiB_{0.6}$ alloy in deaerated 1 N H_2SO_4 solution.

observed following the electrochemical reaction process due to the mass transfer difficulty from the corrosion products of the alloys.[31,32]

Figures 9a and b present experimental and simulated Nyquist plot and Bode plots, respectively, for the $Al_{0.5}CoCrCuFeNi$ alloy in deaerated 1 N H_2SO_4. An equivalent electrical circuit was designed to best fit the experimental results for the electrode and is shown in Fig. 9c, where R_s is the resistance of the solution, R_f and C_f are the resistance and constant phase element associated with the passive film on the surface of the high entropy alloy. The constant phase element, Z_{CPE} is related to the impedance Z, and is given by

$$Z_{CPE} = [Q(j\omega)^n]^{-1} \quad [3]$$

where j is the imaginary root, ω the angular frequency ($\omega = 2\pi f$, f is the frequency), Q the CPE-constant, and n an exponential term. CPEs are used in the analysis of impedance spectra, to account for the deviations caused by surface roughness.[33,34] For a smooth electrode, n approaches unity and the CPE response would be a capacitor. Porous surfaces yield lower n values. R_t is the charge transfer resistance and C_{dl} is the double-layer capacitance, which characterizes the charge separation across the metal and electrolyte interface. When diffusion dominates the process, the $\log|Z|$ vs $\log f$ (frequency) plot will approach a slope of -0.5 and a phase angle of -45 degrees as $\omega \to 0$. Nevertheless, the corrosion process will be partially controlled by the interfacial kinetics, and the slope of $\log|Z|$ against $\log f$ will change with frequency in the range 0 to -0.5, and the phase angle will change with frequency over the range 0 to -45 degrees. Under these conditions, the use of the Warburg impedance (Z_w) is required to fit the data, and is given by[35-37]

$$Z_w = \sigma\omega^{-1/2}(1 - j) \quad [4]$$

where σ is the Warburg coefficient (Ω cm^2 s$^{-1/2}$)

When 0.6 mol of boron was added to the $Al_{0.5}CoCrCuFeNi$ alloy, the Nyquist plot (Fig. 10a) presented two capacitive loops and a diffusion tail at the low-frequency side. Figures 10a and b display the corresponding experimental and simulated Nyquist plot and Bode plots, respectively. The Cr boride precipitates taking place on the surface are the result of the addition of the boron to the $Al_{0.5}CoCrCuFeNi$ alloy. The Cr content on the surface outside the stringy precipitates does not suffice to form a perfect passive film.

Table IV. Equivalent circuit elements values for EIS data corresponding to the $Al_{0.5}CoCrCuFeNi$ (B_0) and $Al_{0.5}CoCrCuFeNiB_{0.6}$ ($B_{0.6}$) alloys in deaerated 1 N H_2SO_4 solution.

	R_S (Ω cm^2)	R_f (Ω cm^2)	C_f (F/cm^2)	n_f	R_{pore} (Ω cm^2)	C_B (F/cm^2)	n_B	R_t (Ω cm^2)	C_{dl} (F/cm^2)	n_{dl}	W (Ω cm^2)
B_0	1.3	1000	2.5×10^{-5}	0.48	—	—	—	2500	7.5×10^{-4}	0.95	1.3×10^{-3}
$B_{0.6}$	1.4	350	6×10^{-5}	0.34	600	1×10^{-3}	0.93	900	8×10^{-3}	0.92	2.5×10^{-2}

The consequence is the localized attacks which appeared on the side of the boride precipitates. It may be concluded that the proposed equivalent electrical circuit to analyze the EIS data is justified as the one shown in Fig. 10c. R_{pore} is the electrolyte resistance through pores,[38,39] and C_B is the capacitance of boride precipitates.

In both situations ($Al_{0.5}CoCrCuFeNi$ and $Al_{0.5}CoCrCuFeNiB_{0.6}$), good agreement between the experimental and the simulated data was established. Table IV gives the simulated values for the equivalent circuit elements. The $Al_{0.5}CoCrCuFeNi$ alloy has higher R_f and R_t values than the $Al_{0.5}CoCrCuFeNiB_{0.6}$ alloy, indicating that high content of boron ($\geqq$0.6 mol) in the $Al_{0.5}CoCrCuFeNiB_x$ alloy impairs the corrosion resistance.

Conclusions

Anodic polarizations demonstrate that the corrosion potential of the $Al_{0.5}CoCrCuFeNi$ high entropy alloy (-0.094 V_{SHE}) is more noble than that of the 304 stainless steel (-0.165 V_{SHE}), and the corrosion current density of the $Al_{0.5}CoCrCuFeNi$ high entropy alloy (3.188×10^{-6} A/cm^2) is lower than that of the 304 stainless steel (3.318×10^{-5} A/cm^2) in deaerated 1 N H_2SO_4 solution. Clearly, the alloy is more resistant to general corrosion than is 304 stainless steel in deaerated 1 N H_2SO_4 solution.

The negative hysteresis in the reverse scans of the cyclic polarization curves indicated that the $Al_{0.5}CoCrCuFeNiB_x$ alloys are not susceptible to localized corrosion in deaerated 1 N H_2SO_4. Furthermore, the corresponding E_{corr} and i_{corr} for the $Al_{0.5}CoCrCuFeNiB_x$ (x = 0, 0.2, 0.6, and 1.0) alloys vary from -115 to -159 mV_{SHE}, and from 787 to 2848 μA/cm^2, respectively.

EDS and XPS analyses reveal that the stringy precipitates of the $Al_{0.5}CoCrCuFeNiB_{0.6}$ alloy are of the Cr, Fe, and Co borides.

Electrochemical impedance spectroscopy indicated that the impedance of the $Al_{0.5}CoCrCuFeNi$ alloy considerably exceeds that of the 304 stainless steel.

The $Al_{0.5}CoCrCuFeNi$ alloy has higher R_f and R_t values than the $Al_{0.5}CoCrCuFeNiB_{0.6}$ alloy, indicating that high content of boron ($\geqq$0.6 mol) in the $Al_{0.5}CoCrCuFeNiB_x$ alloy impairs the corrosion resistance.

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